

Dilan Jayamanne

Software Engineer

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Personal Details

Nationality: Sri Lankan

Visa Status: On Visit Visa / Open to relocation

Age: 25

Marital Status: Single

Availability & Relocation: Immediate

Professional Summary

A results-driven software engineer with 3 years of combined professional experience specializing in high-performance Rust/Python backend development and Rust IoT systems. Proven track record of achieving 60-80% performance improvements through Rust optimization and concurrent programming. Skilled in modern testing practices, profiling, documentation and microservices architecture. Strong advocate for knowledge sharing with 4 years of ICT teaching experience.

Work Experience

Atlas Populi Pvt Ltd

Associate Software Engineer - Level 2 (Individual Contractor)

Sri Lanka (Part time - Remote)

May 2025–Present

- Providing specialized **Rust development consultation** and **code review services** on a part-time basis;
- Supporting development team with advanced Rust programming patterns and **performance optimization**;
- Conducting **pull request reviews** and technical guidance for Rust-based projects.

Atlas Populi Pvt Ltd

Associate Software Engineer - Level 2

Sri Lanka (Full time - Onsite)

Dec 2024–May 2025

- Played an integral role in porting key components to **Rust** for enhanced performance and memory safety, achieving **60-80%** performance improvements on CPU-bound tasks;
- Engineered **Pub/Sub** based live configuration management APIs for edge device updates;
- Worked on a WiFi management service for Linux-based kiosk OS;
- Worked on containerizing microservices and building **CI/CD pipelines**, with containers running on OrangePi ARM-based architecture;
- Worked on stream managers for **HLS live** and **video-on-demand (VOD)** workflows;
- **Documented** and effectively communicated technical solutions to stakeholders;
- Conducted **Rust training** for coworkers, demonstrating advanced knowledge of **ownership, borrowing, and concurrent programming patterns**.

Associate Software Engineer - Level 1

Jan 2024–Dec 2024

- Worked on implementing a **Bluetooth Low Energy (BLE)** service which works both as a peripheral and a central device, streamlining device connectivity and developing reliable solutions for provisioning of IoT devices via smartphones, ensuring both **Android and iOS connectivity** supporting a wider US customer base;
- Researched matter connectivity and QR code provisioning;
- Contributed several Rust scripts for data and backend teams utilizing background tasks using **Redis and cron jobs**;
- Integrated analytics platforms and implemented **telemetry systems** using **Prometheus** to provide real-time monitoring and metrics;
- Worked on scalable REST APIs in Python DRF microservices, **Rust Axum and Hyper frameworks**;
- Created responsive MJML email templates for newsletters and notifications;
- Performed extensive unit testing using **cargo test and profiling for performance optimization**.

Software Engineer Intern (Backend)

Jul 2023–Jan 2024

- Utilized OOP principles and class based views for developing back-end RESTful APIs;
- Enforced customized policies using Python decorators and implemented custom JWT authentication middlewares;
- Conducted R&D on Go Gin framework for authentication middlewares.

Vetstoria

Sri Lanka (Full time - Hybrid)

Software Engineer Intern

Jun 2021–May 2022

- Conducted CI/CD solution evaluation on Bitbucket pipelines;
- Created and maintained API documentation using Swagger and OpenAPI tools;
- Developed and maintained reusable VueJS components, enhancing UI consistency and performance;
- Implemented shell scripts for efficient tracking of pull requests in Bitbucket.

Technical Skills

Core Programming Languages.....

Primary: Rust, Python

Secondary: Java, JavaScript (academic and intern level experience)

Backend Development.....

Frameworks: Axum, Hyper, Django REST Framework, Flask

Databases: PostgreSQL, Redis, MySQL, Firebase RTDB

Authentication: Stateless JWT, API Key, Firebase auth

Testing & Optimization: cargo test unit testing, profiling, python unittest

Specialized Technologies.....

IoT: Bluetooth connectivity and GATT server-client development, shell scripting, video encoding and streaming (HLS), data pipelines

Monitoring: Prometheus instrumenting

Cloud & Services: AWS, Firebase (RTDB, token authentication)

DevOps: Docker, Nerdctl, CI/CD with bitbucket pipelines, Linux ARM compatible container builds

API Documentation: Swagger, OpenAPI

Concurrency: apalis, tokio, dramatiq

Rust Ecosystem: serde, reqwest, bluer, utoipa, zmq, sled, anyhow

Tools & Platforms.....

Version Control: Git

IDEs: VS Code

OS: Development on Linux Ubuntu & Debian ARM architectures runs on OrangePi

Design & Architect tools: Excalidraw, Mermaid charts

Frontend Development.....

Mobile & Web Development: Flutter, React, VueJS (have worked on mini projects)

International Experience & Adaptability

Cross-cultural Work: Atlas Populi has international employees where I engage with them and work on assigned tasks. Worked with an HLS player for React Native where the mobile developer was a remote worker from a different country while I was working on the IoT part with Rust, demonstrating strong cross-cultural collaboration skills.

Research

SyntaxSwap: Supervised Java to JavaScript Code-To-Code Translator

University Final Year Project

2023

- Developed an NLP-based code translation application using GraphCodeBERT with relative positional encoding attention mechanism to convert Java source code to equivalent JavaScript, implemented with PyTorch, Flask, and React.
- **Published:** DOI: 10.13140/RG.2.2.31887.97445
- **Demo video:** <https://youtu.be/xuDMxPEYjs8>

Educational Background

Informatics Institute of Technology

B.Eng. in Software Engineering

Grade: First Class Honours

Affiliated to University of Westminster, UK

Sep 2019–Jul 2023

St. Peter's College

G.C.E. Advanced Level Examination, Commerce stream with ICT

Grade: 3 Distinctions and 1 Very Good Pass

Colombo 04, Negombo branch

May 2017–Aug 2019

Achievements & Competitions

2023: Selected to present final year project (SyntaxSwap) at ICIIT conference and IIT Cutting Edge

2023: Competed up to round 2 in NBQSA presenting final year project

Soft Skills

Leadership: Team work (Led multiple University projects and BLE service @ Atlas Populi), Communication (Proactive bug reporting, IoT solution design, seamless backend-IoT integration), Knowledge sharing

Teaching & Mentoring: Teaching (4 years as self-employed ICT instructor, tutoring programming and developing educational curriculum), Tech Content Creator (Creating educational tech/software engineering videos to support the tech community)

Languages

English: Advanced Professional Proficiency

Sinhala: Native

Interests

Badminton, AI/ML research, technical content creation

References

References available upon request